Problem H2: Electroweak Mixing of $W^{(1,2,3)}$ and B

Glashow–Salam–Weinberg Unification

Parker — PHYS 7502

May 1, 2026

Problem Statement

H2. Give the relation that describes the "mixing" of the "fundamental" W_0 (this means the $\{W_1, W_2, W_3\}$ vector boson triplet) and B vector bosons of the electroweak theory, which get transformed in terms of the weak mixing angle into the usual 4 electroweak bosons we observe (W^+, W^-, Z^0, γ).
(This is related to the "Electroweak Unification" part of Griffiths.)

Deliverables

1. State the gauge group whose structure produces the four "fundamental" bosons $\{W_1, W_2, W_3, B\}$.
2. Give the relation between the charged physical bosons $W_\mu^\pm$ and the $W_\mu^{(1,2)}$ triplet members.
3. Give the (rotation) relation between the neutral physical bosons $\{A_\mu, Z_\mu\}$ and the neutral fundamental bosons $\{W_\mu^{(3)}, B_\mu\}$, expressed in terms of the weak mixing (Weinberg) angle θ_W.
4. Give the definition of θ_W in terms of the two underlying gauge couplings g_W and g', and the consistency relation between g_W, g', and the electromagnetic coupling g_e.
5. Briefly explain *why* the mixing happens (electroweak symmetry breaking).

Background & References

Primary references for this problem.

- Griffiths, *Introduction to Elementary Particles* (2nd ed.), Ch. 10 (§10.6 "Neutral Weak Interactions" and §10.7 "Electroweak Unification"). The mixing matrix is Griffiths' Eq. (10.155); the coupling consistency is Eq. (10.158).
- Thomson, *Modern Particle Physics*, Ch. 15 "Electroweak unification" (§15.2–15.3). The mixing matrix is Eqs. (15.17)–(15.18).
- Halzen & Martin, *Quarks and Leptons*, Ch. 13 ("Weak Interactions of Quarks and Leptons") and Ch. 15 ("Gauge Theories").
- Mann, *An Introduction to Particle Physics and the Standard Model*, Ch. 22 ("Electroweak Unification") and Ch. 23 ("Electroweak Symmetry Breaking"), Eq. (23.45).
- Lancaster & Blundell, *QFT for the Gifted Amateur*, Ch. 47, Eq. (47.35).

Convention note. Griffiths conventions take priority. Be aware that Mann uses the opposite sign convention for $W^\pm$ relative to Griffiths/Thomson. Flagged explicitly below.

Part 1 — Where do $W_{1,2,3}$ and B come from?

Problem Statement

Restatement. Before we can "mix" anything, we have to know where the four fundamental bosons $\{W_1, W_2, W_3, B\}$ come from in the first place. This part identifies the gauge group whose generators give rise to these four gauge fields.

Deliverables

Identify the gauge group of the electroweak interaction and explain how its structure *forces* four gauge bosons (three from $SU(2)_L$, one from $U(1)_Y$).

Strategy

- **Approach 1 (counting):** A non-Abelian gauge symmetry $SU(N)$ has N^2-1 generators, hence N^2-1 gauge bosons. $SU(2)$ gives $2^2-1=3$ gauge bosons. $U(1)$ gives 1. Total $=4$.
- **Approach 2 (analogy with QED):** In QED, the gauge group is $U(1)_{\text{em}}$ and there is one gauge boson, the photon. The electroweak generalization *enlarges* this to $SU(2)_L \otimes U(1)_Y$, with one gauge field per generator.
- **Approach 3 (charges):** Identify the conserved charges. $SU(2)_L$ has weak isospin $\vec{T}$ (3 components); $U(1)_Y$ has weak hypercharge Y (1 component). Gauge bosons couple to charges $\rightarrow$ 4 bosons.

Concept / Definition

The electroweak gauge group is

$$\boxed{G_{\text{EW}} \;=\; SU(2)_L \;\otimes\; U(1)_Y}$$

- $SU(2)_L$ acts only on *left-handed* fermion doublets (subscript L). Its three generators $T^a = \frac{1}{2}\sigma^a$ $(a=1,2,3)$ are the Pauli matrices, and the conserved charge is *weak isospin* $\vec{T}$. Three generators three gauge fields W^a_μ $(a=1,2,3)$, with universal coupling g_W.
- $U(1)_Y$ acts on both chiralities. Its single generator is *weak hypercharge* Y, related to electric charge by the Gell-Mann–Nishijima relation $Q = T^{(3)} + Y/2$. One generator one gauge field B_μ, with coupling g' (Griffiths) or g_Y (Mann).

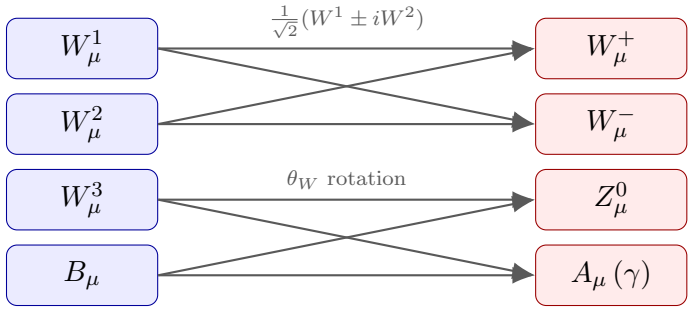
Term Refresher

Weak isospin $\vec{T}$: an internal $SU(2)$ quantum number (no relation to ordinary spin or to strong isospin — Griffiths is emphatic on this point in his footnote on p. 336). Left-handed fermions sit in doublets like $\binom{\nu_L}{e_L}$ with $T^{(3)} = \pm\frac{1}{2}$. Right-handed fermions are $SU(2)_L$ singlets with $T^{(3)} = 0$.
Weak hypercharge Y: the $U(1)_Y$ charge. Defined by $Y = 2(Q - T^{(3)})$ so that the photon, after mixing, couples with the correct electric charge.
Gauge boson: the spin-1 quantum of a gauge field. The number of gauge bosons of a gauge theory equals the number of independent generators of the gauge group.

Physical Insight

We have 3 (from $SU(2)_L$) + 1 (from $U(1)_Y$) = 4 fundamental gauge fields. The observed spectrum also has 4 spin-1 mediators: W^+, W^-, Z^0, γ. The counts match — so the four physical bosons must be linear combinations of $\{W^1, W^2, W^3, B\}$. Finding those linear combinations is exactly what this problem is asking for.

Fundamental ($SU(2)_L \otimes U(1)_Y$) **Physical (mass eigenstates)**



Intuition

Think of $\{W^1, W^2, W^3, B\}$ as the "natural" basis in which the gauge *Lagrangian* is simple to write down (the symmetry is manifest). And $\{W^\pm, Z, \gamma\}$ as the "natural" basis in which the *observed particles* are simple to identify (definite mass and definite electric charge). The mixing relation is just the rotation that takes you between the two bases. The two pairs $\{W^1, W^2\} \to \{W^+, W^-\}$ and $\{W^3, B\} \to \{Z, \gamma\}$ rotate independently of each other.

Part 2 — The charged sector: $W^{(1,2)} \to W^{\pm}$

Problem Statement

Restatement. Express the physical *charged* bosons W_μ^+ and W_μ^- as linear combinations of the gauge fields W_μ^1 and W_μ^2. This part does *not* involve the weak mixing angle — it's pure $SU(2)$ structure.

Deliverables

Write down $W_\mu^\pm$ in terms of $W_\mu^{1,2}$, with the correct sign convention (per Griffiths) and the correct $\frac{1}{\sqrt{2}}$ normalization, and explain where each piece comes from.

Strategy

- **Step 1 — Why pair them up?** The fields W_μ^1 and W_μ^2 are real and electrically neutral as written. But the physical $W^\pm$ carry electric charge $\pm e$, so they cannot be real fields — they must be *complex* (a particle distinct from its antiparticle). The natural way to make a complex field from two real ones is $W^1 \pm iW^2$ up to normalization.
- **Step 2 — Where does the $\frac{1}{\sqrt{2}}$ come from?** It's the normalization that keeps the kinetic term in the Lagrangian canonically normalized, $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$, when we change from real to complex basis. Equivalently, it's the same factor that appears in the $SU(2)$ raising/lowering operators $\sigma_\pm = \frac{1}{2}(\sigma_1 \pm i\sigma_2)$.
- **Step 3 — Sign convention.** Griffiths and Thomson disagree on the sign in front of iW^2. Pick Griffiths' (since the user prefers Griffiths) and flag the alternative.

Step-by-step derivation

(a) Write the $SU(2)_L$ interaction term. The triplet of gauge fields couples to the weak isospin currents as (Griffiths Eq. 10.149)

$$\mathcal{L}_{\text{int}}^{SU(2)} \;=\; -\,g_W\,\vec{j}_\mu \cdot \vec{W}^\mu \;=\; -\,g_W\left(j_\mu^1 W^{1\mu} + j_\mu^2 W^{2\mu} + j_\mu^3 W^{3\mu}\right). \tag{1}$$

Only W^1 and W^2 change weak isospin (they are off-diagonal in the $SU(2)$ doublet basis).

(b) Charge raising/lowering. The combinations of weak-isospin generators that *raise* or *lower* weak isospin are

$$\sigma_\pm \;=\; \tfrac{1}{2}\left(\sigma_1 \pm i\sigma_2\right), \qquad \sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},\; \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{2}$$

σ_+ takes $e_L \to \nu_L$ (raises $T^{(3)}$ by 1), and the Hermitian conjugate field that mediates this transition must be W^+ (a charged $+e$ boson is emitted/absorbed).

(c) Define the physical charged fields. By analogy with $\sigma_\pm$, define

$$\boxed{W_\mu^\pm \;\equiv\; \frac{1}{\sqrt{2}}\left(W_\mu^1 \pm i\,W_\mu^2\right) \quad \text{[Griffiths Eq. (10.152) convention]}} \tag{3}$$

Equivalently $W_\mu^1 = \frac{1}{\sqrt{2}}(W_\mu^+ + W_\mu^-)$, $W_\mu^2 = \frac{i}{\sqrt{2}}(W_\mu^+ - W_\mu^-)$.

(d) Verify that the Lagrangian becomes simple. Substitute into the charged piece $j^1 W^1 + j^2 W^2$ and use $j^\pm = j^1 \mp i j^2$ (the corresponding current combinations). After a one-line algebra, one gets

$$j^1_\mu W^{1\mu} + j^2_\mu W^{2\mu} \;=\; \tfrac{1}{\sqrt{2}} \left(j^+_\mu W^{-\mu} + j^-_\mu W^{+\mu} \right), \tag{4}$$

which is exactly the structure that gives the standard $V-A$ charged-current vertex $\bar{u}\gamma^\mu \frac{1}{2}(1-\gamma^5)d \cdot W^+_\mu$ (Griffiths Eq. (10.153)–(10.154)).

Warning / Convention Pitfall

Sign convention pitfall. Two common conventions exist:

- **Griffiths** (Eq. 10.152): $W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \pm i W^2_\mu)$.
- **Thomson** (Eq. 15.12) and **Mann** (Eq. 23.45): $W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp i W^2_\mu)$.

The two differ only in which combination is labeled "+" versus "−." Whichever convention you pick must be applied consistently with the corresponding sign choice for $\sigma_\pm$ and $j^\pm$. *Don't mix conventions across textbooks.*

For Prof. Frantz's class, since the rest of the chapter follows Griffiths, stay with Griffiths' convention.

Key Result

$$\boxed{W^\pm_\mu \;=\; \frac{1}{\sqrt{2}} \left(W^1_\mu \pm i\, W^2_\mu \right)}$$

This is just a unitary change of basis: two real fields → one complex field and its conjugate. No mixing angle yet — the charged sector is fully fixed by $SU(2)$ structure alone.

Part 3 — The neutral sector: $\{W^3, B\} \to \{\gamma, Z\}$

Problem Statement

Restatement. Now do the same for the two *neutral* fundamental bosons W^3_μ and B_μ. Show that they get rotated by an angle θ_W (the weak mixing angle / Weinberg angle) into the photon A_μ and the Z boson Z_μ. *This is the heart of the question.*

Deliverables

1. Argue why W^3 *cannot be* the photon all by itself (and why B *cannot be* the photon all by itself).
2. Write the 2×2 rotation between $\{W^3, B\}$ and $\{\gamma, Z\}$ using Griffiths' sign convention.
3. Display the inverse relation $\{W^3, B\}$ in terms of $\{\gamma, Z\}$.

Strategy

- **Step 1 — Why mix at all?** Argue from physics: the photon couples *equally* to left- and right-handed fermions (proven by parity-conserving QED), but W^3 couples only to left-handed fermions. Therefore $W^3 \neq \gamma$. By the same token B couples to weak hypercharge Y, not electric charge Q, so $B \neq \gamma$. Both are needed.
- **Step 2 — The most general mixing.** The most general transformation between two real, neutral fields that preserves the kinetic-term normalization is an $SO(2)$ rotation, i.e. a 2×2 orthogonal matrix parameterized by a single angle θ_W.
- **Step 3 — Fix the angle.** The angle is fixed by the requirement that one linear combination (the photon) couples with strength g_e to electric charge Q, not to weak isospin or weak hypercharge separately. This is what *electroweak unification* means at the technical level.
- **Step 4 — Write down the matrix.** Use Griffiths Eq. (10.155).

Why the mixing has to happen — the physical argument

Physical Insight

The photon is the gauge boson of $U(1)_{\text{em}}$, the symmetry that survives *after* the Higgs vacuum breaks $SU(2)_L \otimes U(1)_Y \to U(1)_{\text{em}}$. The unbroken $U(1)_{\text{em}}$ generator is a specific linear combination of the broken-symmetry generators:

$$Q \;=\; T^{(3)} \;+\; \tfrac{1}{2}Y. \qquad \text{(Gell-Mann–Nishijima relation)} \tag{5}$$

Because Q is a *linear combination* of $T^{(3)}$ and Y, the gauge boson A_μ that couples to Q must likewise be a linear combination of the gauge bosons that couple to $T^{(3)}$ (which is W^3_μ) and Y (which is B_μ). The orthogonal combination is the Z^0.

The mixing matrix

The most general orthogonal 2×2 rotation is

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix}. \tag{6}$$

Reading off rows:

$$\boxed{\begin{aligned} A_\mu &= B_\mu \cos\theta_W + W_\mu^3 \sin\theta_W, \\ Z_\mu &= -B_\mu \sin\theta_W + W_\mu^3 \cos\theta_W. \end{aligned}} \quad \text{[Griffiths Eq. (10.155); Thomson Eqs. (15.17)–(15.18)]} \tag{7}$$

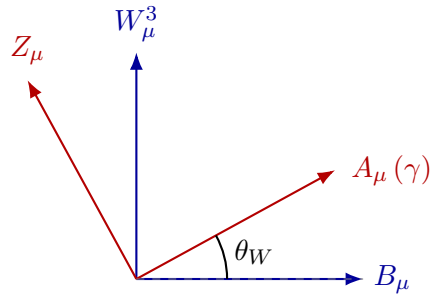
The inverse is found by transposing the rotation matrix (orthogonal, so $R^{-1} = R^T$):

$$\begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix}, \qquad \begin{aligned} B_\mu &= A_\mu \cos\theta_W - Z_\mu \sin\theta_W, \\ W_\mu^3 &= A_\mu \sin\theta_W + Z_\mu \cos\theta_W. \end{aligned} \tag{8}$$

Warning / Convention Pitfall

Convention check. Mann (Eq. 23.45) writes the inverse as $W_\mu^3 = \cos\theta_W\, Z_\mu - \sin\theta_W\, A_\mu$ and $B_\mu = \sin\theta_W\, Z_\mu + \cos\theta_W\, A_\mu$. This corresponds to a rotation by $-\theta_W$ relative to Griffiths — i.e., the sign of θ_W is flipped. The physical content (the mixing angle and the structure of the rotation) is identical; only the labeling direction is reversed. Griffiths' convention is the more common one in modern textbooks (Thomson, Halzen & Martin, Lancaster & Blundell all agree with Griffiths).

Visualizing the rotation



The physical photon A_μ and Z_μ are obtained from the fundamental B_μ and W_μ^3 by a rotation through $\theta_W \approx 28.7°$.

Intuition

The picture is exactly the same kind of rotation you saw with neutrino oscillations (flavor states vs. mass states) and CKM mixing (weak eigenstates vs. mass eigenstates). In each case, the Lagrangian is most naturally written in one basis (the *interaction* or *symmetry* basis) but the physical particles — those with definite mass — live in another (the *mass* basis). The rotation between them encodes physics; the *angle* is a measured parameter of the Standard Model, not predicted from first principles.

Key Result

The full mixing relation (the answer to H2):

$$\boxed{\begin{aligned} W_\mu^\pm &= \tfrac{1}{\sqrt{2}}\,(W_\mu^1 \pm i\,W_\mu^2), \\ A_\mu &= \quad B_\mu \cos\theta_W + W_\mu^3 \sin\theta_W \quad (=\gamma), \\ Z_\mu &= -B_\mu \sin\theta_W + W_\mu^3 \cos\theta_W . \end{aligned}}$$

Equivalently, the four physical fields $\{W^+, W^-, Z, A\}$ are obtained from the four fundamental fields $\{W^1, W^2, W^3, B\}$ by a unitary transformation that decouples into (charged sector 2×2) $\oplus$ (neutral sector 2×2).

Part 4 — The Weinberg angle and electroweak coupling consistency

Problem Statement

Restatement. The mixing angle θ_W is not a free parameter once we demand that the photon A_μ couple with strength g_e to electric charge Q. Show how θ_W is fixed in terms of the underlying gauge couplings g_W (the $SU(2)_L$ coupling) and g' (the $U(1)_Y$ coupling), and write the consistency relation between g_W, g', and g_e.

Deliverables

1. Define θ_W in terms of g_W and g'.
2. State the electroweak unification consistency relation $g_W \sin\theta_W = g' \cos\theta_W = g_e$.
3. Quote the experimentally measured value of $\sin^2\theta_W$.

Strategy

- **Step 1 — Substitute the mixing relation into the gauge interaction Lagrangian.** The interaction term in the symmetry basis is

$$\mathcal{L} \supset -g_W\, j^3_\mu W^{3\mu} - \tfrac{g'}{2}\, j^Y_\mu B^\mu.$$

  Use the inverse mixing $W^3 = A\sin\theta_W + Z\cos\theta_W$, $B = A\cos\theta_W - Z\sin\theta_W$ to re-express in terms of A and Z.
- **Step 2 — Read off the photon coupling.** The piece that multiplies A_μ must equal $-g_e\, j^{\text{em}}_\mu$, where $j^{\text{em}}_\mu = \sum_f Q_f\, \bar\psi_f \gamma_\mu \psi_f$ is the standard QED current.
- **Step 3 — Match.** Use the Gell-Mann–Nishijima relation $j^{\text{em}} = j^3 + \frac{1}{2} j^Y$ to identify coefficients. Out drops the famous double-equality $g_W \sin\theta_W = g' \cos\theta_W = g_e$, which fixes $\tan\theta_W = g'/g_W$.

Step-by-step derivation

(a) Substitute into the Lagrangian. Start from the neutral piece

$$\mathcal{L}_{\text{neut}} = -g_W\, j^3_\mu W^{3\mu} - \tfrac{g'}{2}\, j^Y_\mu B^\mu. \tag{9}$$

Using $W^{3\mu} = A^\mu \sin\theta_W + Z^\mu \cos\theta_W$ and $B^\mu = A^\mu \cos\theta_W - Z^\mu \sin\theta_W$:

$$\begin{aligned}\mathcal{L}_{\text{neut}} &= -g_W\, j^3_\mu (A^\mu \sin\theta_W + Z^\mu \cos\theta_W) - \tfrac{g'}{2}\, j^Y_\mu (A^\mu \cos\theta_W - Z^\mu \sin\theta_W) \qquad (10)\\ &= -A^\mu \left(g_W \sin\theta_W\, j^3_\mu + \tfrac{g'}{2} \cos\theta_W\, j^Y_\mu \right) \\ &\quad - Z^\mu \left(g_W \cos\theta_W\, j^3_\mu - \tfrac{g'}{2} \sin\theta_W\, j^Y_\mu \right). \qquad (11)\end{aligned}$$

(b) Demand QED form for the photon. The QED interaction reads $\mathcal{L}_{\text{QED}} = -g_e\, j^{\text{em}}_\mu A^\mu$, with $j^{\text{em}}_\mu = j^3_\mu + \frac{1}{2} j^Y_\mu$ (Gell-Mann–Nishijima at the current level). Comparing the A_μ piece in (11):

$$g_W \sin\theta_W\, j^3_\mu + \tfrac{g'}{2} \cos\theta_W\, j^Y_\mu \overset{!}{=} g_e \left(j^3_\mu + \tfrac{1}{2} j^Y_\mu \right). \tag{12}$$

Matching coefficients of j^3_μ and j^Y_μ separately:

$$[j^3]: \qquad g_W \sin\theta_W = g_e, \tag{13}$$

$$[j^Y]: \qquad g' \cos\theta_W = g_e. \tag{14}$$

(c) The two consistency equations together.

$$\boxed{g_W \sin\theta_W = g' \cos\theta_W = g_e} \qquad \text{[Griffiths Eq. (10.158); Thomson Eq. (15.30)]} \tag{15}$$

(d) Solve for θ_W. Dividing the two equations:

$$\boxed{\tan\theta_W = \frac{g'}{g_W}.} \tag{16}$$

Solving for the trig functions individually using $\sin^2 + \cos^2 = 1$:

$$\sin\theta_W = \frac{g'}{\sqrt{g_W^2 + g'^2}}, \qquad \cos\theta_W = \frac{g_W}{\sqrt{g_W^2 + g'^2}}, \qquad g_e = \frac{g_W\, g'}{\sqrt{g_W^2 + g'^2}}. \tag{17}$$

(e) The Z^0 coupling drops out as a bonus. Reading the Z^μ piece of (11) and using the consistency relations,

$$\mathcal{L}_Z = -Z^\mu\left(g_W \cos\theta_W\, j^3_\mu - \tfrac{g'}{2}\sin\theta_W\, j^Y_\mu\right) = -\frac{g_e}{\sin\theta_W\cos\theta_W}\, Z^\mu\left(j^3_\mu - \sin^2\theta_W\, j^{\text{em}}_\mu\right), \tag{18}$$

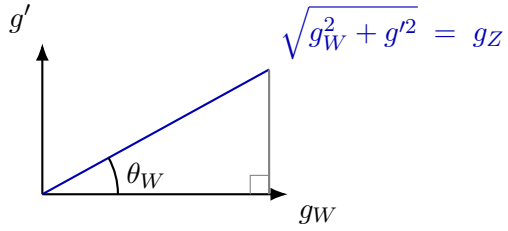
which is Griffiths Eq. (10.159). The combination $g_e/(\sin\theta_W\cos\theta_W) \equiv g_Z$ is what Griffiths denotes by g_z in his Eq. (10.94).

Concept / Definition

The geometry of θ_W. If you draw a right triangle with legs of length g_W ("horizontal") and g' ("vertical"), the angle that the hypotenuse makes with the g_W axis is θ_W:

$$\tan\theta_W = \frac{g'}{g_W}, \qquad \sin\theta_W = \frac{g'}{\sqrt{g_W^2 + g'^2}}, \qquad \cos\theta_W = \frac{g_W}{\sqrt{g_W^2 + g'^2}}.$$

The hypotenuse length $\sqrt{g_W^2 + g'^2}$ is related to the Z-boson coupling $g_Z = g_W/\cos\theta_W = \sqrt{g_W^2 + g'^2}$.



Experimental value

The Weinberg angle is *not* predicted by the GWS model — it is an input parameter to be measured. Modern global fits (PDG/LEP/SLD/Tevatron/LHC averages) give:

$$\sin^2\theta_W \approx 0.23146 \pm 0.00012 \qquad \text{(Thomson Eq. 15.31)}, \tag{19}$$

which corresponds to $\theta_W \approx 28.75°$. Griffiths quotes the same value (his Eq. 10.95) to slightly less precision, $\sin^2\theta_W \approx 0.23$, $\theta_W \approx 28.7°$.

Physical Insight

The fact that $\sin^2\theta_W$ is *neither* 0 *nor* 1 is what makes the electroweak theory *electroweak* — the photon and the Z are genuine *mixtures* of W^3 and B, not just one or the other. Specifically:

- If $\theta_W = 0$: photon $= B$ exactly, $Z = W^3$ exactly — no unification, just two unrelated $U(1)$'s.
- If $\theta_W = 90°$: photon $= W^3$, $Z = -B$ — nonsensical (photon would couple only to left-handed fields).
- The measured $\theta_W \approx 28.7°$ says: the photon is mostly B ($\cos^2\theta_W \approx 77\%$) with a substantial W^3 admixture ($\sin^2\theta_W \approx 23\%$). The Z is the orthogonal combination.

Key Result

The Weinberg angle:

$$\tan\theta_W = \frac{g'}{g_W}, \qquad g_W \sin\theta_W = g' \cos\theta_W = g_e, \qquad \sin^2\theta_W \approx 0.231.$$

This single parameter ties together *three* couplings (g_W, g', g_e) and *four* gauge fields ($W^{1,2,3}$, $B \to W^{\pm}, Z, \gamma$). That is the precise sense in which electromagnetism and the weak force are "unified."

Part 5 — Why does the mixing happen? Symmetry breaking

Problem Statement

Restatement. Conceptually: *why* are the four "fundamental" bosons $\{W^{1,2,3}, B\}$ *not* themselves the observed bosons? What physical mechanism forces the rotation? Briefly explain the role of the Higgs field.

Deliverables

Give a one-paragraph physical explanation: the Higgs vacuum expectation value (VEV) spontaneously breaks $SU(2)_L \otimes U(1)_Y \to U(1)_{\text{em}}$, giving masses to three linear combinations of the gauge fields ($W^{\pm}, Z$) and leaving one combination (γ) massless. The mixing angle is determined by the Higgs's hypercharge.

Strategy

- **Symmetry argument:** Three broken generators three Goldstone bosons three gauge fields "eat" them and become massive. The unbroken generator's gauge boson stays massless.
- **Mass-matrix argument (more concrete):** Compute the gauge-field mass matrix from $|D_\mu \Phi|^2$ at $\langle \Phi \rangle = (0, v/\sqrt{2})^T$. The mass matrix is 2×2 in the (W^3, B) subspace and is *not* diagonal — the off-diagonal entries are exactly what forces the θ_W rotation.
- **Punchline:** the eigenvector of the mass matrix with eigenvalue 0 is the photon; the eigenvector with eigenvalue $\frac{v^2}{4}(g_W^2 + g'^2)$ is the Z. The angle is $\tan\theta_W = g'/g_W$ — exactly what we found in Part 4.

Sketch of the argument (Halzen & Martin Ch. 15; Mann Ch. 23; Lancaster & Blundell Ch. 47)

(a) Higgs Lagrangian. The Higgs scalar Φ is an $SU(2)_L$ doublet with hypercharge $Y_\Phi = +1$:

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad v \approx 246 \text{ GeV}. \tag{20}$$

(b) Covariant derivative. The Higgs kinetic term in the Lagrangian is $|D_\mu \Phi|^2$, with

$$D_\mu \Phi = \left(\partial_\mu - i g_W T^a W_\mu^a - i \tfrac{g'}{2} Y_\Phi B_\mu \right) \Phi. \tag{21}$$

Plug in $\langle \Phi \rangle$. The spontaneous symmetry breaking generates mass-like terms quadratic in the gauge fields:

$$|D_\mu \langle \Phi \rangle|^2 \supset \frac{v^2}{8} \left[g_W^2 (W_\mu^1)^2 + g_W^2 (W_\mu^2)^2 + (g_W W_\mu^3 - g' B_\mu)^2 \right]. \tag{22}$$

The first two terms give masses to W^1, W^2 (combine into $W^\pm$ with mass $m_W = g_W v/2$). The third term is a 2×2 quadratic form in (W^3, B):

$$\frac{v^2}{8} (W_\mu^3, B_\mu) \begin{pmatrix} g_W^2 & -g_W g' \\ -g_W g' & g'^2 \end{pmatrix} \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}. \tag{23}$$

(c) Diagonalize the (W^3, B) mass matrix. The eigenvalues of $\begin{pmatrix} g_W^2 & -g_W g' \\ -g_W g' & g'^2 \end{pmatrix}$ are

$$\lambda_+ = g_W^2 + g'^2, \qquad \lambda_- = 0. \tag{24}$$

The eigenvector for λ_+ is $\propto (g_W, -g')$, which (after normalizing) is exactly $Z_\mu = -B_\mu \sin\theta_W + W_\mu^3 \cos\theta_W$ with $\tan\theta_W = g'/g_W$. The eigenvector for $\lambda_- = 0$ is $\propto (g', g_W)$, which is $A_\mu = B_\mu \cos\theta_W + W_\mu^3 \sin\theta_W$. **This is the same rotation we wrote down in Part 3.**

(d) Masses. Reading off:

$$m_W = \tfrac{1}{2} g_W v, \qquad m_Z = \tfrac{1}{2} v \sqrt{g_W^2 + g'^2} = \frac{m_W}{\cos\theta_W}, \qquad m_\gamma = 0. \tag{25}$$

The relation $m_W = m_Z \cos\theta_W$ is one of the most precisely-tested predictions of the Standard Model (Griffiths Eq. 10.98; Thomson §16.4).

Physical Insight

The mixing angle θ_W is *not* a free input of GWS at the deepest level — it is determined by the ratio of the two gauge couplings g'/g_W. What *is* free in the GWS model is the choice of g_W and g' themselves. Equivalently, you can pick any two of $\{g_e, g_W, g', \theta_W\}$ as the independent inputs.

The Higgs mechanism does not directly *predict* θ_W; it just guarantees that the mass eigenstates are obtained from the symmetry eigenstates by an $SO(2)$ rotation parameterized by exactly $\tan\theta_W = g'/g_W$. This is why the Weinberg angle shows up *geometrically* in the mass matrix diagonalization.

Intuition

In one sentence: The Higgs vacuum is electrically neutral but carries weak isospin and weak hypercharge. So it "feels" three of the four electroweak generators (specifically T^1, T^2, and the combination $T^{(3)} - \frac{1}{2}Y$) and gives mass to the corresponding gauge fields. The fourth generator ($Q = T^{(3)} + \frac{1}{2}Y$) *annihilates* the vacuum, so the corresponding gauge field stays massless — and *that* is the photon.

Summary — the answer to H2 in one box

Key Result

The electroweak gauge group is $SU(2)_L \otimes U(1)_Y$, with four fundamental gauge fields:

$$\underbrace{W_\mu^1,\, W_\mu^2,\, W_\mu^3}_{SU(2)_L \text{ triplet}} \qquad \text{and} \qquad \underbrace{B_\mu}_{U(1)_Y \text{ singlet}} \quad .$$

The four physical bosons (mass eigenstates) are obtained by a unitary change of basis:

$$\boxed{\begin{aligned} W_\mu^\pm &= \tfrac{1}{\sqrt{2}}\left(W_\mu^1 \pm i\, W_\mu^2\right), \\ A_\mu &= \;\; B_\mu \cos\theta_W \;+\; W_\mu^3 \sin\theta_W, \\ Z_\mu &= -B_\mu \sin\theta_W \;+\; W_\mu^3 \cos\theta_W, \end{aligned}}$$

where the *Weinberg angle* θ_W is defined by

$$\tan\theta_W \;=\; \frac{g'}{g_W}, \qquad g_W \sin\theta_W \;=\; g' \cos\theta_W \;=\; g_e \quad \text{(electroweak unification)}.$$

Experimental value: $\sin^2\theta_W \approx 0.231$, i.e. $\theta_W \approx 28.7°$.

Why the mixing is forced: the Higgs VEV breaks $SU(2)_L \otimes U(1)_Y \to U(1)_{\text{em}}$. The mass matrix in the (W^3, B) subspace has off-diagonal entries proportional to $g_W g'$, so the mass eigenstates (γ and Z) are rotations of the symmetry eigenstates. The γ is the massless eigenvector; Z is the massive orthogonal combination. The rotation angle is θ_W.

One-line cheat sheet

$$W^\pm = \tfrac{1}{\sqrt{2}}(W^1 \pm iW^2), \quad \begin{pmatrix} A \\ Z \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B \\ W^3 \end{pmatrix}, \quad \tan\theta_W = g'/g_W.$$